GarbageX

Abstract :

Solid waste that is floating in the water surface is detrimental to the environment and is a risk to the wildlife, in this we discuss a simple device to remove solid waste from the surface of the water bodies such as the lake, river and the sea. We will look at a device that uses the properties of adhesiveness to remove the waste in an effective manner. The waste is collected in a container and can be disposed of in a proper way. The device can be attached to any pre existing vessels.

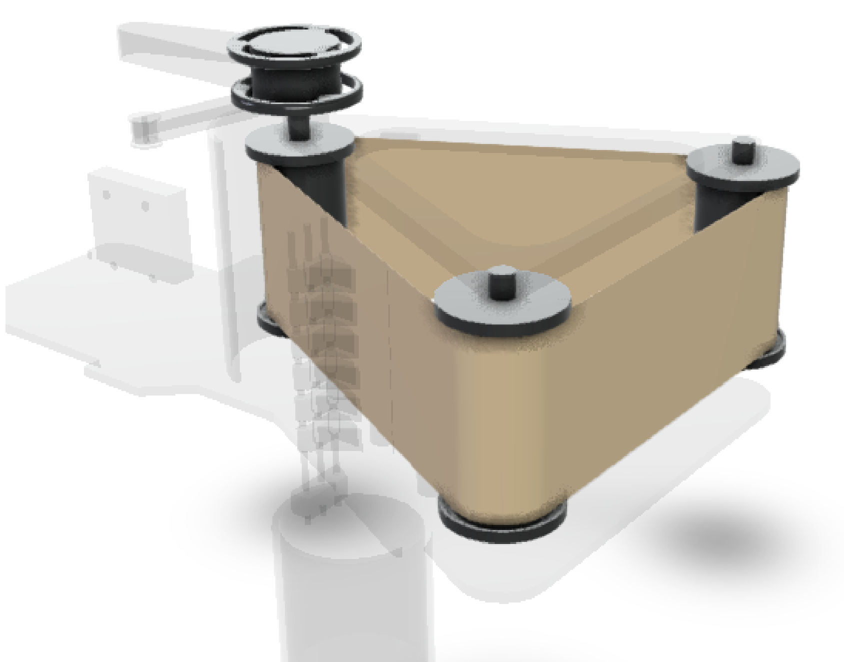
Introduction :

GarbageX is used to remove the solid wastes that are floating on the surface of the water bodies. It uses the waterproof adhesive property of certain compounds to effectively collect the particles to a conveyer adhesive belt, then with use of a scraper the stuck particles are scraped off into waste bin attached to the bottom of the setup.

Tool: Fusion 360

Design:

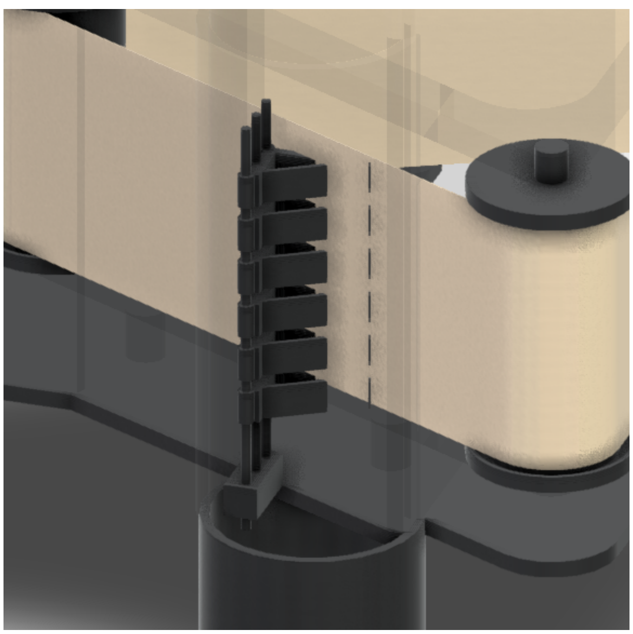
The main component of the device is a belt which has a coating of a special waterproof adhesive. The belt moves along three vertical rollers which are powered by an electric motor via a leather belt, only two rollers face the water surface which creates



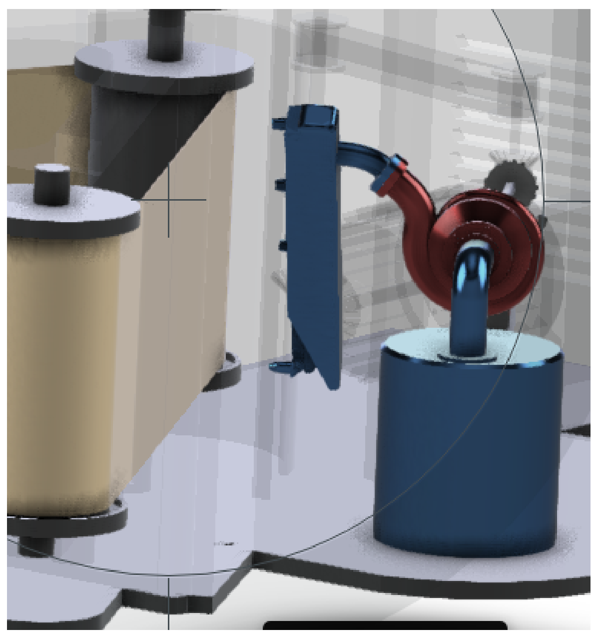
a flat surface for the belt to attach to the solid wastes. The wastes are removed from the belt with the help of a scrapper which removes the wastes from the belt without removing the adhesive, but there is a certain loss of adhesive which occurs over time, to maintain the operation of the device a sprayer is used to replenish the adhesive. The waste from the belt is collected in a container for disposal.

Working :

The entire device is capable of being attached to any ship or vessel with the help of mounting points in the frame. The device is then driven with the belts facing forward. The belt is driven from left to right with the help of an electric motor. When the belt makes contact with the wastes floating on the surface it sticks with it and pulls it along the rollers. The wastes are removed with the help of a scrapper, it scrapes the surface of the belt to remove the wastes and collects the waste in a container located below. The belt is then routed back to the water surface for repeating the process. During the scraping process some of the adhesive may be removed which reduces the operating efficiency of the device so a adhesive pump with a nozzle is fitted to the belt to replenish the adhesive.



Scraper



Adhesive sprayer

Conventional methods :

The main methods of removing the solid wastes from the water bodies was by using modified fishing nets which was attached to a boat or other vehicle and the waste is collected through capturing and is emptied to a contained but the problem with this method is the water bodies contain waste material that varies in size so if a larger net is used, smaller sized objects will slip through and the device will be ineffective and if a smaller finer net is used it is more susceptible to tearing due to the stress induced by larger objects. So two different size nets of two devices with different size nets have to be used to clean the same surface which is inefficient. With the help of adhesives larger and smaller objects can be removed since the adhesive property will work on smaller waste material and larger objects won't do damage to the belt and since the larger object has more surface area, it will stick to the belt and can be removed. Unlike the nets, GarbageX is easy to maintain, demands very low manpower and compact.

Energy regeneration system :

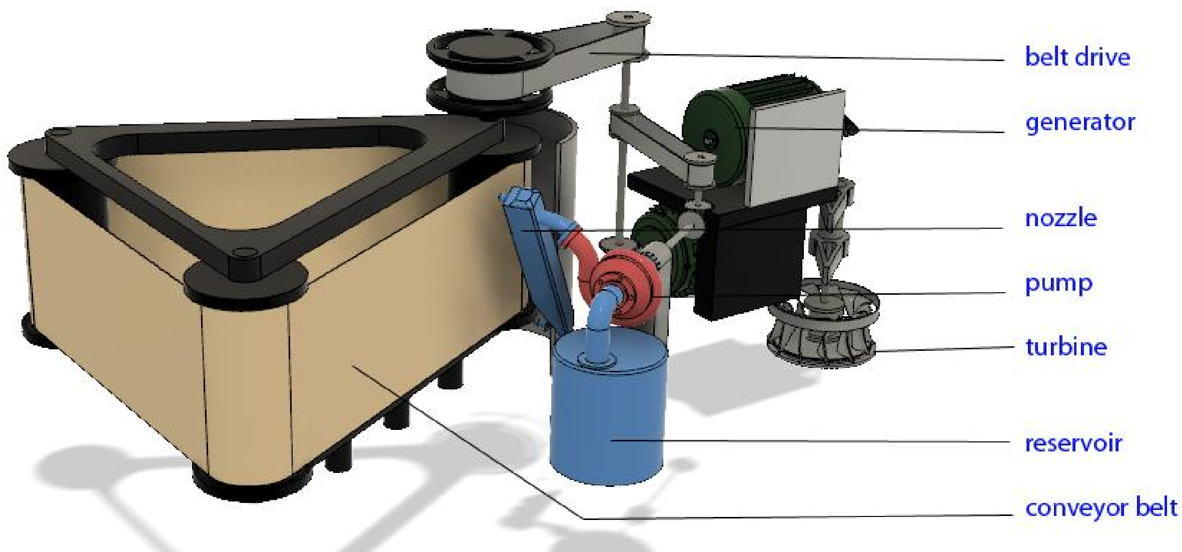
The bottom of the device houses a turbine which is connected to a generator, when the device is moving a water current is induced and the turbine spins to produce electricity, this electricity is harvested and stored in an onboard battery. The excess energy is then used to run the electric motor for the functioning of the device.

Advantages :

- The actual working part of the device experienced less wear and tear as the belt is very simple in construction.
- The belt does not oxidise or wears out in normal operating conditions
- The conveyor system is more robust than the traditional systems because of less point of failure.
- The adhesive works on a wide range of size of the wastes.

- This design requires less maintenance.

Parts:



Conclusion :

The solid waste in the water bodies proves to be damaging for the environment. The adhesive solid waste remover is a design that is primarily focused on effectiveness and longevity. It can be deployed in a variety of situations and has wide operating conditions. Using this device we can clean the water bodies which are infested with solid wastes.

Source link: https://a360.co/2Re1Amb (For Design View)

Renders :